

Anway Pimpalkar

PhD Student and NDSEG Fellow at Harvard University

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EDUCATION

Harvard University	2025 –
PhD, Engineering Sciences: Materials Science & Mechanical Engineering	
Secondary Field in Mind, Brain and Behaviour	
Advisors: Katia Bertoldi, PhD and Giovanni Traverso, PhD	
Johns Hopkins University	2023 – 2025
MSE, Biomedical Engineering	
Thesis: Bridging Perception & Performance: Generating, Calibrating, & Evaluating Multimodal Sensory Feedback	
Advisor: Jeremy D. Brown, PhD	
College of Engineering Pune, India	2019 – 2023
BTech, Electronics and Telecommunication Engineering	

AWARDS & HONORS

Harvard School of Engineering and Applied Sciences Prize Fellowship • Harvard University	2025 – 2031
\$281,796 • 6 yr tuition, 2 yr stipend	
National Defense Science & Engineering Graduate Fellowship • US DOW, Office of Naval Research	2025 – 2028
\$277,484 • 3 yr tuition, 3 yr stipend	
GEM Associate Fellowship • National GEM Consortium	2025 – 2031
Best Student Paper Finalist [for C2] • International Conference on Rehabilitation Robotics (ICORR)	2025
NIA Start-Up Challenge Finalist • National Institute on Aging of the NIH	2025
\$10,000 • Article	
Best Paper Award [for C1] • International Conference on Biomedical Engineering Science and Tech.	2023
Mitacs Globalink Research Award • Dalhousie University, Canada	2022
CA \$9,915	
Best Project Award [for O1] • IEEE National Project Competition, India	2021

RESEARCH EXPERIENCE

Harvard University and MIT • Bertoldi Lab , Lab for Translational Engineering	Cambridge, MA
Doctoral Research Fellow • Advisors: Katia Bertoldi, PhD and Giovanni Traverso, PhD	
Oct 2025 –	
- Multistable mechanisms and adaptive structures for interactive human-device interfaces - Mechanical systems and assistive robotics to support cognitive and physical functions	
Johns Hopkins University • Haptics and Medical Robotics Lab	Baltimore, MD
Graduate Research Assistant • Advisor: Jeremy D. Brown, PhD	
Jan 2024 – Jun 2025	
- Multimodal visual-haptic neurorehabilitation for improving post-stroke finger dexterity - Pneumatactors: Soft multimodal haptic interfaces	
Harvard University • Biohybrid Organs and Neuroprosthetics Lab	Cambridge, MA
Research Fellow • Advisor: Shriya S. Srinivasan, PhD	
May – Aug 2024	
- Peripheral nerve stimulation to modulate cancer tumor metastasis - Surgical approach for treating stress urinary incontinence	

- IIT Bombay • Human Motor Neurophysiology and Neuromodulation Lab** [↗](#) Mumbai, India
 Research Intern • Advisor: Nivethida Thirugnanasambandam, PhD Jan – Aug 2023
- Quantifying prospective component of sense of agency using machine learning
- College of Engineering Pune • Department of Applied Sciences** [↗](#) Pune, India
 Undergraduate Researcher • Advisor: Mahesh H. Shindikar, PhD Aug 2022 – Aug 2023
- Deep learning-based MRI skull tissue segmentation
- Dalhousie University • Agricultural Mechanized Systems Group** [↗](#) Truro, Canada
 Mitacs Globalink Research Intern • Advisor: Travis J. Esau, PhD May – Aug 2022
- Automating harvesters to reduce operator cognitive load
- Queliz Lifetech • Early Stage Medtech Startup** Pune, India
 Research Intern, Electronics Subsystem Jun – Sep 2021
- Point-of-care device for finger dexterity rehabilitation in acute burn patients

JOURNAL PUBLICATIONS

- J4 **Pimpalkar A**, Brown JD. *Pneumatactors: Soft 3D-printed interface for co-located transient and continuous tactile stimuli*. In preparation. 2026.
- J3 **Pimpalkar A**, West AM, Rai D, Xu J, Brown JD. *Perceptually calibrated haptic feedback provides limited benefit beyond vision in a virtual precision grip task*. In preparation. 2026.
- J2 Harris C, **Pimpalkar A**, Aggarwal A, Yang J, Chen X, Schmidgall S, Rapuri S, Greenstein JL, Taylor CO, Stevens RD. *Preoperative risk prediction of major cardiovascular events in noncardiac surgery using the 12-lead electrocardiogram: explainable deep learning approach*. Br J Anaesth. 2025 Sep; doi: 10.1016/j.bja.2025.07.085 [↗](#)
 Featured in *The Hub at Johns Hopkins*. [Article](#) [↗](#)
- J1 **Pimpalkar A**, Slepian A, Thakor NV. *Vibrations at first contact encode object stiffness before grasp completion*. IEEE Sens Lett. 2025 Aug;9(8):1–4. Doi: 10.1109/lens.2025.3584860 [↗](#)
 Featured in *New Scientist Magazine*. [Article](#) [↗](#)

CONFERENCE PRESENTATIONS [†](#) presenter

FULL-LENGTH PAPERS

- C5 West AM, Kasimcan B, **Pimpalkar A**, Xu J, Brown JD. *Cross-modal matching is not bidirectional: Implications for human multisensory feedback*. 2026 IEEE RAS EMBS Int Conf Biomed Robot Biomechatron (BioRob), Sep 2026. In print. [Poster Pres]
- C4 Salgado AS, Bartlett JS, **Pimpalkar A**, Brown JD, Reed KB [†](#). *Teaching at the Teachable Moment Using Asynchronous Videos Combined with Live Interactions*. 2026 ASEE Annual Conference & Exposition. Jun 2026. In print. [Oral Pres]
- C3 West AM [†](#), **Pimpalkar A**, Xu J, Brown JD. *SenseMatch: Smartphone-based cross-modal matching for accessible perceptual assessments*. 2025 Int IEEE EMBS Conf Neural Eng (NER), 2025 Nov. p. 85–90. Doi: 10.1109/ner61569.2025.11589033 [↗](#) [Spotlight Poster Pres]
- C2 **Pimpalkar A** [†](#), West AM, Xu J, Brown JD. *Optimizing cross-modal matching for multimodal motor rehabilitation*. 2025 IEEE Int Conf Rehabil Robot (ICORR). 2025 May. P. 559–566. Doi: 10.1109/icorr66766.2025.11063112 [↗](#) [Podium Pres] **Best Student Paper Award Finalist**.
- C1 **Pimpalkar A** [†](#), Patole RK, Kamble KD, Shindikar MH. *Performance evaluation of vanilla, residual, and dense 2D U-Net architectures for skull stripping of augmented 3D T1-weighted MRI head scans*. 2023 Int Conf Biomed Eng Sci Tech. 2023 Feb. p. 131–142. Doi: 10.1007/978-3-031-54547-4_11 [↗](#) [Podium Pres] **Best Paper Award**.

ABSTRACTS AND SHORT PAPERS

- A8 West AM [†](#), Kasimcan B, **Pimpalkar A**, Xu J, Brown JD. *Cross-modal matching is not bidirectionally invertible: Implications for multisensory representation*. 2026 Neural Control of Movement Meet. 2026 Apr. [Poster Pres]

- A7 **Pimpalkar A** †, Rai D, Bartels UJ, Xu J, Brown JD. *Visual-haptic feedback enhances finger individuation in a virtual precision grip neurotraining task*. 2024 Biomed Eng Soc Ann Meet (BMES). 2024 Oct. [Podium Pres, [Abs ↗](#)]
- A6 Song H † et al. [including **Pimpalkar A**]. *Towards visual function restoration through photoacoustic stimulation*. Invest Ophthalmol Vis Sci. 2024 Jun. [Podium Pres, [Abs ↗](#)]
- A5 **Pimpalkar A** †, Ameta P †, Dalia A, Brown JD. *Pneumatactor arrays for high frequency vibrotactile feedback*. 2024 IEEE Haptics Symp. 2024 Apr. [Live Demo, Work-in-Prog Paper]
- A4 **Pimpalkar A** †, Ameta P †, Dalia A †, Brown JD. *Pneumatactor arrays for high frequency vibrotactile feedback*. 2024 US Senate AI Cauc Robotics Demo Day. 2024 Apr. [Live Demo]
- A3 Harris C †, **Pimpalkar A**, Aggarwal A, Yang P, Chen X, Taylor CO, Greenstein J, Stevens RD. *Surgical risk prediction using an explainable deep learning approach applied to pre-operative 12-lead electrocardiograms*. 2024 JH Med & Eng Res Retr. 2024 Feb. [Poster Pres]
- A2 **Pimpalkar A** †, Patole R, Thirugnanasambandam N. *Demonstrating the prospective component of sense of agency using machine learning*. 2023 COEP ENT C Btech Proj Symp. 2023 May. [Poster Pres]
- A1 **Pimpalkar A** †, Patole R, Kamble K, Shindikar M. *Evaluating U-Nets for skull stripping of augmented T1-weighted MRI scans*. 2023 No Garland Neurosci Conf. 2023 Feb. [Podium Pres]

NON PEER-REVIEWED ARTICLES

- O1 **Pimpalkar A**, Niture D. *Towards contactless elevators with 3inyML using person detection and keyword spotting*. 2024 Jul. doi: arXiv:2405.13051 [↗](#) **Best Project Award**.

TEACHING ENGAGEMENTS

Principles of the Design of Biomedical Instrumentation • Johns Hopkins University 580.771 Fall 2024

Teaching Assistant to Nitish V. Thakor, PhD

- Mentored labs on designing ECG, EMG, EEG circuits and projects including accessible gaming, wearables.

Haptic Interface Design for Human-Robot Interaction • Johns Hopkins University 530.691 Fall 2024

Class Researcher with Jeremy D. Brown, PhD

- Helped test an online learning platform providing self-paced lectures incorporating live instructor interaction.
- Resulted in co-authored manuscript presentation to 2026 ASEE Annual Conference [[C4](#)].

Mechatronics • Johns Hopkins University 530.421 Spring 2024

Teaching Assistant to Jeremy D. Brown, PhD

- Solved a 3-year challenge in real-time location polling of autonomous air hockey robots using CV-based ArUco tag detection and ZigBee mesh networking. Led labs on building autonomous robots. [GitHub ↗](#)

MENTORING

UNDERGRADUATES

George Sealy • Summer Intern from University of Cambridge, Materials Science Jul – Sep 2026

Project: Mechanics and acoustics of elastic structures.

Jocelyn Hwang • Summer Intern from MIT, Electrical Engineering and Computer Science Jun – Sep 2026

Project: Multiplexing electronics for multi-electrode biosensing.

Fuxing (Fu) Chen • Undergraduate Researcher from MIT, Mechanical Engineering Jan 2026 –

Project: Elastic actuators for ingestible robots and drug delivery.

PRESENTATIONS AND INVITED LECTURES

T2 **Indian Institute of Technology Bombay** • Human Motor Neurophysiology & Neuromodulation Lab Jun 2025

T1 **Johns Hopkins University** • Neural & Rehabilitation Engineering Course Sep 2024

ACADEMIC SERVICE

IEEE Haptics Symposium • Student Volunteer	2024
Canadian Society of Bioengineering Annual Meeting • Student Volunteer	2022

IN MEDIA

Johns Hopkins University Hub ↗ <i>"AI fares better than doctors at predicting deadly complications after surgery"</i>	Sep 17, 2025
New Scientist ↗ <i>"Delicate robot hands know just how hard to squeeze"</i>	Jan 2, 2025
Johns Hopkins University Hub ↗ <i>"A game-changing gait lab in a shoe"</i>	May 9, 2025
Arduino Blog ↗ <i>"This contactless system combines embedded ML and sensors to improve elevator safety"</i>	Jan 29, 2022

LEADERSHIP AND TEAMWORK

Johns Hopkins India Institute • Operations Team • Managed on-day operations and logistics for the first annual Hopkins-India Conference.	2024 – 2025
Johns Hopkins Biomedical Engineering • Master's Program Representative	2024 – 2025
COEP Impressions Cultural Festival • Head of Events (2021), Volunteer • Led the planning and execution of 28 annual events with 10k attendees, managing a ₹2m budget and coordinating with an overall team of 126 members to execute concerts, competitions, and international celebrity engagements. • Built and directly led a team of 15, working with 2 co-heads to build a platform from the ground up.	2019 – 2021
Rowing Team • COEP, Pune City • Competitive 1X, 2X, 4X Sculler • Won race events and an institutional award commending skill and enthusiasm. Represented city at regional races.	2020 – 2022
COEP Mental Health & Wellness Center • Student Volunteer	2020 – 2022
COEP Data Science & AI Club • Research Lead (2020-21), Member	2020 – 2022